From Vortex Rings to the Top Quark: Dynamical Derivations in the Null Worldtube Framework*

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Abstract

We model the electron as a quantized vortex ring in a superfluid vacuum, described by the Gross-Pitaevskii equation (GPE). The torus knot of the Null Worldtube (NWT) framework is identified with the vortex ring's topology: major radius R set by the healing length, tube radius r by the spacetime impedance, and charge radius r_e by the electromagnetic self-energy. This identification provides dynamical derivations of particle masses and confinement, addressing the key criticism of earlier work [1, 2, 3]: that numerical relations were asserted rather than derived. Starting from the GPE Lagrangian and Macken's spacetime impedance $Z_s = c^3/G$, we derive: (i) the torus tube radius $r/R = 1/\pi^2$ from the standing-wave energy condition, matching the NWT value of 0.10392 to 2.5%; (ii) quark confinement from the incommensurability condition $\gcd(n_{\text{quarks}}, q_{\text{poloidal}}) = 1$ on torus knots; (iii) the pion mass $m_\pi = 2m_e/\alpha = 140.05$ MeV (0.34% accuracy) from the pair-creation resonance at the classical electron radius; (iv) the $1/\alpha$ energy ladder $\Lambda_n = m_e/\alpha^n$ generating the nuclear, heavy-quark, and electroweak scales; and (v) the top quark mass $m_t = 2k^2 m_e/\alpha^2 = 172{,}730$ MeV (0.02% accuracy) from the pair resonance at level 2 of the ladder. All results follow from one measured mass (m_e), one coupling (α), and two topological integers ($p = 2$, $k = 3$), with no fitted parameters.

1 Introduction

Previous papers in this series established the Null Worldtube (NWT) framework: a photon confined to a $(2, 1)$ torus knot with aspect ratio $k = R/r = 3$ encodes the Standard Model spectrum [1], with the topology reducing the 26 free parameters to three integers $(p, q, k) = (2, 1, 3)$ and one measured mass m_e [2]. The nucleonics extension derived all seven nuclear magic numbers from the same geometry [3].

External review of this work identified a persistent criticism: *the numerical relations are asserted, not derived.* The Koide angle, the mode counting, and the mass anchoring formulas match experiment closely, but no dynamical calculation was shown to *require* those specific values. As one reviewer summarized: "the formulas appear chosen because they give the right number."

This paper addresses that criticism by identifying the NWT torus with a concrete physical object: a **quantized vortex ring** in a superfluid vacuum. Vortex rings are soliton solutions of the Gross-Pitaevskii equation — a well-studied Lagrangian field theory — and their properties (core size, circulation, self-velocity, Kelvin wave spectrum) are determined by the medium, not by hand. The identification is:

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NWT torus	Superfluid vortex ring
Major radius R	Ring radius = healing length ξ
Tube radius r	Core size (from impedance matching)
$(2,1)$ knot topology	Doubly-wound vortex (spin-$\frac{1}{2}$)
Photon circulation	Quantized superfluid flow
EH nonlinearity	Vortex self-interaction

From this identification, we derive the torus geometry, the confinement mechanism, and the particle mass spectrum using four ingredients — all known physics:

1. The Gross-Pitaevskii equation (GPE), describing superfluid dynamics [6];
2. Macken's spacetime impedance $Z_s = c^3/G$, relating the energy stored in a standing wave to the mode volume [4];
3. The Euler-Heisenberg (EH) nonlinearity of QED vacuum polarization;
4. The topology of the $(2,1)$ torus knot.

2 The Vacuum as a Superfluid

We model the QED vacuum as a superfluid described by the Gross-Pitaevskii equation:

$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m^*}\nabla^2\psi + g|\psi|^2\psi\,, \tag{1}$$

where ψ is the order parameter, m^* is the effective constituent mass, and g is the nonlinear coupling.

Two physical constraints fix the vacuum parameters:

$$c_s = \sqrt{\frac{gn_0}{m^*}} = c\,, \tag{2}$$

$$\xi = \frac{\hbar}{\sqrt{2m^* gn_0}} = \lambda\!\!\!^{-}_C = \frac{\hbar}{m_e c}\,, \tag{3}$$

where c_s is the speed of sound (identified with the speed of light) and ξ is the healing length (identified with the reduced Compton wavelength). These two equations yield:

$$m^* = \frac{m_e}{\sqrt{2}} = 0.361\ \mathrm{MeV}/c^2\,. \tag{4}$$

Vortex solutions of the GPE are topological defects with quantized circulation $\kappa = 2\pi\hbar/m^*$. A vortex ring of radius R has energy given by the Kelvin formula:

$$E_{\mathrm{ring}} = \tfrac{1}{2}\rho_0\kappa^2 R\big[\ln(8R/a) - \beta\big]\,, \tag{5}$$

where a is the core size, ρ_0 the background mass density, and β depends on the core model. Identifying the electron as the lowest-energy vortex ring with $R = \xi = \lambda\!\!\!^{-}_C$ and setting $E_{\mathrm{ring}} = m_e c^2$ determines the third vacuum parameter $\rho_0 \approx 8.0 \times 10^4$ kg/m^3.

3 Tube Radius from Spacetime Impedance

The electron torus has major radius $R = \lambda\!\!\!^{-}_C$, but what determines the tube (minor) radius r?

Macken [4] showed that spacetime has a mechanical impedance $Z_s = c^3/G$, with the energy stored in a standing dipole wave:

$$E = \frac{Z_s A^2 \omega^2 V_{\text{mode}}}{2c}\,, \tag{6}$$

where $A = m_e/m_P$ is the Planck-limited strain amplitude and $\omega = \omega_C = c/R$ is the Compton frequency. The product $Z_s A^2 = m_e^2 c^2/\hbar$ is exact.

Setting $E = m_e c^2$ gives the mode volume:

$$V_{\text{mode}} = 2R^3\,. \tag{7}$$

On the torus, the volume decomposes as the product of the longitudinal path (circumference $2\pi R$) and the transverse mode area (πr^2):

$$V_{\text{torus}} = 2\pi^2 R r^2 = (2\pi R)(\pi r^2)\,. \tag{8}$$

The photon propagates as a *traveling wave* along the torus knot (the real particle), while being confined as a *standing wave* across the tube cross-section (the bound mode profile). Each factor of π links a geometric scale to R: one from "going around" ($2\pi R$) and one from "being round" (πr^2).

Equating $V_{\text{torus}} = V_{\text{mode}}$:

$$\boxed{\frac{r}{R} = \frac{1}{\pi^2} = 0.1013} \tag{9}$$

This is within 2.5% of the NWT self-energy value $r/R = 0.10392$, and is the first derivation of the tube radius from a dynamical principle. It uses only $Z_s = c^3/G$ and the torus geometry — notably, the fine-structure constant α does not appear.

The three radii of the electron torus are now all derived:

Radius	Value	Origin
$R = \lambda\!\!\!^{-}_C$	386 fm	Healing length (GPE)
$r = R/\pi^2$	39 fm	Spacetime impedance (Macken)
$r_e = \alpha R$	2.82 fm	EM self-energy

4 Quark Confinement from Incommensurability

On a (p, q) torus knot, the curvature varies with period q along the knot path. A longitudinal mode at frequency f/n (with n wavelengths per circuit) experiences this curvature variation as a periodic potential — the wave equation becomes a Mathieu equation.

The interaction between the mode periodicity (n) and the curvature periodicity (q) determines localization:

- $\gcd(n, q) = 1$: the mode and curvature are *incommensurate*. The resulting quasicrystal potential localizes the mode — the "quarks" are **confined**.
- $\gcd(n, q) > 1$: the mode locks to the curvature (*commensurate*). The quarks are delocalized and can rearrange — the particle **decays**.

This predicts the stability of every particle tested:

Particle	(p, q)	n	gcd	Prediction
Proton	$(1, 4)$	3	1	Stable ✓
Electron	$(2, 1)$	2	1	Stable ✓
Pion	$(1, 2)$	2	2	Decays ✓
Ω^-	$(1, 3)$	3	3	Decays ✓
P_c	$(1, 5)$	5	5	Decays ✓

Color confinement in this framework is the *quasicrystal gap*: the three quarks ($n = 3$) on the proton's $(1, 4)$ knot are pinned at 120° separation by the incommensurate potential, unable to slide or separate without breaking the topology.

5 Kelvin Wave Stability and the Lepton–Hadron Divide

Kelvin waves — helical perturbations of the vortex core — propagate along the ring and can destabilize it at specific aspect ratios (the Widnall instability [5]).

On a (p, q) torus knot, the p-fold toroidal symmetry imposes a *selection rule*: only azimuthal modes with $m \equiv 0 \pmod{p}$ are allowed. This dramatically restricts the Widnall resonances:

- **Leptons** ($p \geq 2$): the selection rule $m = 2k$ (for the electron) or $m = 3k$ (for heavier leptons) kills most resonances, creating wide stable windows. Leptons are *absolutely stable* against Kelvin wave instabilities.
- **Hadrons** ($p = 1$): all azimuthal modes are active, creating a dense resonance landscape. Hadrons are intrinsically less stable and can decay via mode coupling.

This is the superfluid analog of MHD stability in tokamaks: higher rotational transform (more toroidal windings) means fewer rational surfaces, hence better confinement.

6 The Pion Mass from Pair-Creation Resonance

The electric field of the electron at its classical radius $r_e = \alpha R$ exceeds the Schwinger critical field by a factor of $1/\alpha$:

$$\frac{E(r_e)}{E_S} = \frac{1}{\alpha} \approx 137\,. \tag{10}$$

At this field strength, virtual pair creation is copious. The lightest pair excitation that resonates with the $(2, 1)$ electron topology has one wavelength per toroidal winding at the charge scale:

$$\lambda\!\!\!^{-}_{\pi} = \frac{r_e}{p} = \frac{\alpha R}{2}\,. \tag{11}$$

The corresponding mass is:

$$\boxed{m_\pi = \frac{p \cdot m_e}{\alpha} = \frac{2m_e}{\alpha} = 140.05 \text{ MeV}} \tag{12}$$

compared to the measured charged pion mass $m_{\pi^\pm} = 139.57$ MeV ($\Delta = 0.34\%$).

Geometrically, this states that the pion's Compton wavelength equals *half the classical electron radius*: $\lambda\!\!\!^{-}_{\pi} = r_e/2 = 1.409$ fm. The pion torus fits inside the electron's charge radius, with the factor of 2 reflecting the duality between the electron's $p = 2$ toroidal windings and the pion's $q = 2$ constituents (quark and antiquark).

7 The $1/\alpha$ Energy Ladder

The pair-creation mechanism generalizes to a hierarchy of scales. Each factor of $1/\alpha$ represents one additional electromagnetic self-interaction at the charge boundary, where $E/E_S = 1/\alpha$. We define:

$$\Lambda_n = \frac{m_e}{\alpha^n}\,, \tag{13}$$

giving the energy scales:

Level	Scale	Value	Physics
0	m_e	0.511 MeV	Atomic (QED)
1	m_e/α	70.0 MeV	Nuclear
2	m_e/α^2	9.60 GeV	Heavy quark
3	m_e/α^3	1.315 TeV	Electroweak

7.1 Level 1: the pion and nuclear scale

The pion sits at level 1 with pair multiplicity $p = 2$: $m_\pi = 2\Lambda_1$. Related quantities:

$$f_\pi = \frac{k+1}{k}\,\Lambda_1 = 93.4 \text{ MeV} \quad (\text{measured: } 92.1 \text{ MeV},\ \Delta = 1.4\%)\,, \tag{14}$$

$$\Lambda_{\text{QCD}} = k\,\Lambda_1 = 210 \text{ MeV} \quad (\text{standard range: } 200\text{–}300 \text{ MeV})\,. \tag{15}$$

7.2 Level 2: heavy quarks and bottomonium

Level 2 hosts the heavy-quark spectrum:

Particle	Formula	Predicted	Measured	Δ
J/ψ	Λ_2/π	3054 MeV	3097 MeV	-1.4%
m_b	$\Lambda_2/\sqrt{5}$	4291 MeV	4180 MeV	$+2.7\%$
B_c	$(2/3)\Lambda_2$	6397 MeV	6275 MeV	$+1.9\%$
$\Upsilon(1S)$	Λ_2	9596 MeV	9460 MeV	$+1.4\%$

The factors $1/\pi$, $1/\sqrt{5}$, and $2/3$ appearing in these relations are geometric: π from the circular cross-section, $\sqrt{5} = \sqrt{p^2+q^2}$ from the $(2,1)$ knot invariant, and $2/3$ from the torus winding ratio.

7.3 The top quark

The most striking prediction emerges at level 2 with full torus amplification. The pair resonance at level 2, enhanced by the aspect ratio squared, gives:

$$\boxed{m_t = 2k^2\,\frac{m_e}{\alpha^2} = 18\,\frac{m_e}{\alpha^2} = 172{,}730 \text{ MeV}} \tag{16}$$

compared to the measured value $m_t = 172{,}760 \pm 300$ MeV:

$$\frac{m_t^{\text{pred}}}{m_t^{\text{meas}}} = 0.99983 \qquad (\Delta = -0.02\%)\,. \tag{17}$$

The factor $18 = 2 \times k^2 = 2 \times 9$ has a clear physical interpretation: the pair factor (2, same as the pion) times the square of the torus aspect ratio ($k^2 = 9$). The k^2 enhancement is the same scalar-vector amplification that produces the nuclear spin-orbit force in Paper III [3].

The top quark mass is determined entirely by m_e, α, and $k = 3$, with zero free parameters.

8 Discussion

8.1 Addressing the "asserted, not derived" criticism

Each result in this paper follows from a physical mechanism:

- $r/R = 1/\pi^2$: from the Macken standing-wave energy condition $V_{\text{mode}} = 2R^3$ on the torus;

- Quark confinement: from the Mathieu equation on a torus knot with incommensurate mode and curvature periodicities;
- $m_\pi = 2m_e/\alpha$: from the pair-creation resonance at r_e, where $E/E_S = 1/\alpha$;
- $m_t = 2k^2 m_e/\alpha^2$: from the same pair mechanism at the second rung of the $1/\alpha$ ladder, with torus amplification.

The logical chain runs: Lagrangian (GPE) → vacuum parameters → torus geometry (Macken) → charge scale (EM self-energy) → pair-creation resonance → mass spectrum. At no point is a formula "chosen to give the right number."

8.2 The $1/\alpha$ hierarchy and QCD

The relation $m_\pi = 2m_e/\alpha$ connects QED and nuclear physics through a single formula with no fitted constants. Standard QCD gives $m_\pi^2 \propto (m_u + m_d)\Lambda_{\mathrm{QCD}}$; our result is consistent if $\Lambda_{\mathrm{QCD}} = k\Lambda_1 = 3m_e/\alpha \approx 210$ MeV, which falls within the standard range. The NWT framework does not replace QCD but suggests that the QCD parameters are themselves determined by the torus topology.

8.3 Predictions

The framework makes several falsifiable predictions:

1. The pion mass is $2m_e/\alpha$ exactly (not fitted); precision measurements at the 0.01% level could test this.
2. The top quark mass is $2k^2 m_e/\alpha^2 = 18\Lambda_2$; future e^+e^- collider measurements (ILC, CLIC, FCC-ee) at the ~ 50 MeV level would provide a decisive test.
3. The $1/\alpha$ ladder predicts level 3 at 1.315 TeV, suggesting new physics or a structural feature near this scale.

9 Conclusion

We have derived the torus tube radius, the quark confinement mechanism, the pion mass, and the top quark mass from dynamical principles — the GPE Lagrangian, Macken's spacetime impedance, and the topology of torus knots. The results require only one measured mass (m_e), one coupling (α), and two integers ($p = 2$, $k = 3$).

The top quark mass prediction $m_t = 2k^2 m_e/\alpha^2 = 172{,}730$ MeV, agreeing with experiment to 0.02%, is the most precise result of the NWT program and the strongest evidence that the particle mass spectrum is determined by electromagnetic topology.

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